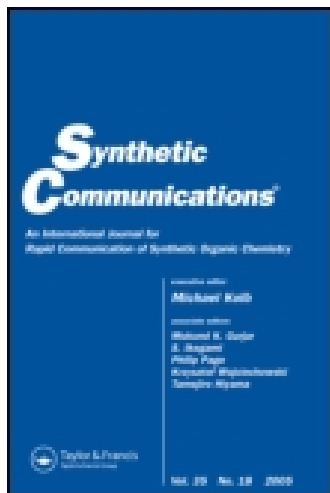




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Diastereoselective Addition of Organozinc Reagents to Benzaldiminetricarbonylchromium Derivatives

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ABSTRACT

The reaction of benzaldiminetricarbonylchromium derivatives with benzylzinc bromide or allylzinc bromide in the presence of $\text{BF}_3 \cdot \text{Et}_2\text{O}$ gave the corresponding amine complexes with high diastereoselectivities.

Key Words: Imines; Addition; Chromium.

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Nucleophilic addition to planar chiral tricarbonyl(1,2- or 1,3-disubstituted arene)chromium complexes has been an attractive approach for synthesis of natural products^[1] because the bulkiness of tricarbonylchromium moiety plays an important role for promoting the reaction with quite high facial selectivities, and the tricarbonylchromium moiety is easily removed by air and sunlight. In recent years, the chiral chromium complexes have been used for synthesis of important skeletons such as (–)-lasubine(I),^[2] mitosanes,^[3] (*R*)-(–)-baclofen,^[4] vancomycin,^[5] 11-*epi*-helioporin B,^[6] and fredericamycin A.^[7] We have recently reported that novel reaction of benzaldiminetricarbonylchromium derivatives with 2-silyloxy-1,3-butadienes in the presence of zinc triflate and water gave Mannich-type products with high stereoselectivities.^[8] In spite of the development of stereoselective addition to the benzaldiminetricarbonylchromium derivatives,^[9] nucleophiles have been quite limited compared to those in the reactions of aldimines, and we observed that stronger reaction conditions were necessary in the reaction of the chromium complexes.^[10]

In this communication, we describe diastereoselective addition of organozinc reagents to benzaldiminetricarbonylchromium derivatives in the presence of Lewis acids. Organozinc reagents have been considerable attention since the introduction of highly reactive zinc.^[11] Recently Buttero et al. reported that the reaction of the chromium complexes with α -bromoester in the presence of zinc by using ultrasound irradiation afforded β -aminoesters or β -lactams.^[12] We thought that the reaction of the chromium complexes with organozinc bromide simply by using Lewis acid would afford the corresponding amines with high diastereoselectivities.

Benzaldiminetricarbonylchromium complexes **1** were prepared as previously described.^[13] First, the reaction of the chromium complex **1a** with Grignard reagent (1.2 equiv. to **1a**) was carried out (Table 1). In the case of benzylmagnesium chloride, desired product was not observed by ¹H NMR even though some Lewis acids were used. However, the use of benzylmagnesium bromide without Lewis acid gave the corresponding amine complex **2a** with >97% diastereoselectivity which was determined by ¹H NMR. Removal of tricarbonylchromium moiety of the crude chromium product **2a** by air and sunlight in dichloromethane afforded the corresponding amine **3a**, which is well known as one of isoquinoline precursors,^[14] in 85% isolated yield after column chromatography.

Next we examined the reactivities of organozinc reagents to the aldimine complexes (Table 2). The reaction of the complex **1a** with benzylzinc bromide without Lewis acid did not give the product, but the addition of BF₃·Et₂O to the complex afforded the desired amine complex



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Table 1.

Lewis acid	Grignard reagent	de of 2a (%) ^a	Yield of 3a (%) ^b
—	BnMgCl	—	0
ZnCl ₂	BnMgCl	—	0
Zn(OTf) ₂	BnMgCl	—	0
Sc(OTf) ₃	BnMgCl	—	0
—	BnMgBr	> 97	85

To a stirred solution of chromium complex (0.25 mmol) in THF (5 mL) was added Lewis acid (0.5 mmol) and grignard reagent (0.75 mmol) at 0°C. The mixture was stirred at r.t. for 12 h.

^aDe was measured by ¹H NMR.

^bIsolated yield after chromatography.

Table 2.

Entry	R ¹	R ²	R ³	Complex	R ⁴ ZnBr	Lewis acid	de of 2 (%) ^a	Yield of 3 (%) ^b
1	Me	H	<i>o</i> -anisyl	1a	BnZnBr	—	—	< 20 (3a)
2	Me	H	<i>o</i> -anisyl	1a	BnZnBr	BF ₃ ·Et ₂ O	> 97	74 (3a)
3	H	Me	<i>o</i> -anisyl	1b	BnZnBr	BF ₃ ·Et ₂ O	62	51 (3b)
4	MeO	H	<i>o</i> -anisyl	1c	BnZnBr	BF ₃ ·Et ₂ O	> 97	80 (3c)
5	Me	H	<i>o</i> -anisyl	1d	allylZnBr	BF ₃ ·Et ₂ O	> 97	87 (3d)

To a stirred solution of chromium complex (0.25 mmol) in THF (5 mL) was added BF₃·Et₂O (0.5 mmol) and organozinc reagent (0.75 mmol) at r.t. The mixture was stirred at 30°C for 6 h.

^aDe was measured by ¹H NMR.

^bIsolated yield after chromatography.



with high diastereoselectivity (Entry 2). Removal of tricarbonylchromium moiety was carried out as described above, and the corresponding amine **3a** was obtained in 74% yield after column chromatography. Under similar conditions, reactions of aldimine complexes **1b–c** were carried out. High diastereoselectivities were observed except for the reaction with the complex having *m*-tolyl substituent **1b** in which rotation between carbon atom of C=N bond and α -carbon of aldimine might occur. The reaction with allylzinc bromide also afforded the corresponding amine complex diastereoselectively under similar conditions (Entry 5). From the results, high diastereoselectivities in the reaction of organozinc reagents were observed as in the case with Grignard reagents. In addition, benzylzinc bromide in THF is stable on storage compared to the corresponding Grignard reagent.

In summary, the reaction of the chromium complexes with organozinc reagents in the presence of Lewis acid gave the corresponding amine complexes with high diastereoselectivities. Application to natural products by using planar chiral benzaldiminetricarbonylchromium complexes is now in progress.

EXPERIMENTAL

Typical Experimental Procedure for Reaction of Chromium Complexes with Organozinc Reagents

To a stirred solution of chromium complex **1** (0.25 mmol) and $\text{BF}_3 \cdot \text{Et}_2\text{O}$ (0.5 mmol) in dry THF (5 mL) was added organozinc reagent (0.75 mmol) at room temperature. The reaction mixture was stirred at 30°C for 6 h in the dark, and was quenched with aqueous NaHCO_3 . The mixture was extracted with ether (20 mL $\times$ 3), and the combined organic layers were dried over Na_2SO_4 . The precipitates were filtered off, and the solvent was evaporated to give the crude chromium product. The de was determined by ^1H NMR. To the crude chromium product was added dichloromethane (60 mL) and the mixture was exposed to air and sunlight for 1 h. The solvent was evaporated and ether was added to the mixture. The precipitates were filtered off, and the solvent was evaporated to give the crude amine which was purified by flash column chromatography on SiO_2 (*n*-hexane/EtOAc = 3/1). Isolated yields were measured after the chromatography.

Amine 3a. ^1H NMR (500 MHz, CDCl_3): δ 2.39 (s, 3H), 3.00–3.05 (m, 1H), 3.14 (dd, 1H, J = 14.1, 5.0 Hz), 3.84 (s, 3H), 4.76–4.79 (m, 1H), 6.59–6.75 (m, 3H), 7.18–7.49 (m, 10H). HRMS-FAB(M^+): Obsd m/z



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317.1786. Calcd. for $C_{22}H_{23}NO$: 317.1780. Anal. calcd. for $C_{22}H_{23}NO$: C, 83.24; H, 7.30; N, 4.41. Found: C, 83.49; H, 7.32; N, 4.40.

Amine 3b. 1H NMR (500 MHz, $CDCl_3$); δ 1.45 (s, 3H), 2.85–2.86 (m, 1H), 3.00–3.02 (m, 1H), 3.79 (s, 3H), 4.90–4.91 (m, 1H), 6.48–6.82 (m, 3H), 7.10–7.17 (m, 10H). HRMS-FAB(M^+): Obsd m/z 317.1777. Calcd. for $C_{22}H_{23}NO$: 317.1780. Anal. calcd. for $C_{22}H_{23}NO$: C, 83.24; H, 7.30; N, 4.41. Found: C, 83.56; H, 7.52; N, 4.49.

Amine 3c. 1H NMR (500 MHz, $CDCl_3$); δ 2.84–2.88 (m, 1H), 2.94 (dd, 1H, $J=13.4$, 5.0 Hz), 3.79 (s, 3H), 3.83–3.85 (m, 1H), 4.67 (s, 3H), 6.79–7.38 (m, 13H). HRMS-FAB(M^+): Obsd m/z 333.1736. Calcd. for $C_{22}H_{23}NO_2$: 333.1729. Anal. calcd. for $C_{22}H_{23}NO_2$: C, 79.25; H, 6.95; N, 4.20. Found: C, 79.46; H, 6.82; N, 4.08.

Amine 3d. 1H NMR (500 MHz, $CDCl_3$); δ 2.38 (s, 3H), 3.80 (s, 3H), 4.50 (dd, 1H, $J=8.1$, 4.9 Hz), 5.09 (d, 1H, $J=11.8$ Hz), 5.17 (d, 1H, $J=11.8$ Hz), 5.68–5.81 (m, 1H), 6.08 (d, 1H, $J=7.7$ Hz), 6.48–6.69 (m, 3H), 7.03–7.17 (m, 5H), 7.30–7.33 (m, 1H). HRMS-FAB(M^+): Obsd m/z 267.1631. Calcd. for $C_{18}H_{21}NO$: 267.1623. Anal. calcd. for $C_{18}H_{21}NO$: C, 80.86; H, 7.92; N, 5.24. Found: C, 80.72; H, 8.22; N, 5.65.

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